Metadata prediction from epigenomic data applied to over 43,000 public datasets

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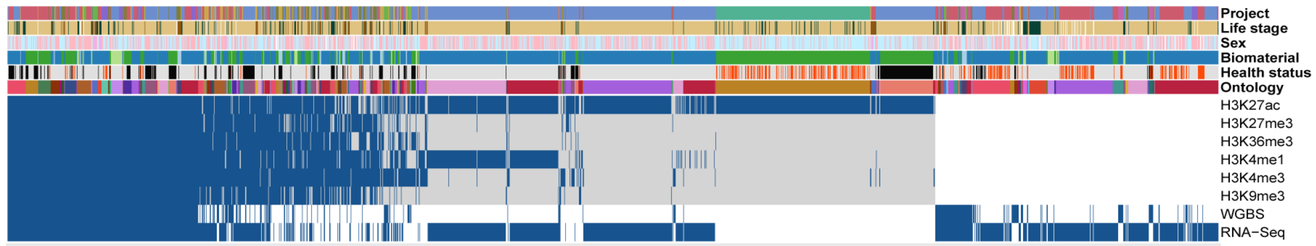
Abstract

Epigenomic data found in public databases often suffer from issues of non-standardization and incompleteness in their associated metadata. To tackle this challenge, we harnessed the extensive and harmonized data and metadata provided by the EpiAtlas project of the International Human Epigenome Consortium (IHEC). Using this rich resource, we trained highly accurate classifiers capable of predicting important metadata information, such as the conducted assays and characteristics of the biological samples, including sex, cell type, biomaterial type, and life stage. Orthogonal analyses supported the predictions of the classifiers. The development of these classifiers enabled the identification of a few mislabeled and low-quality datasets in EpiAtlas, while also completing the vast majority of the missing metadata. The characterization of the most relevant genomic regions for the classifiers confirmed that they are biologically relevant.

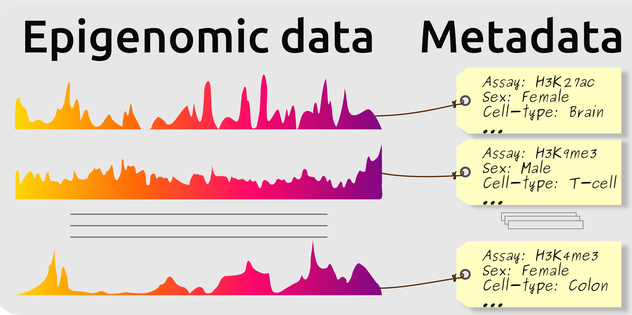
These classifiers were next applied to assess more than 43,000 datasets from public repositories. This effort not only validated the accuracy of the majority of assays reported by the original authors, but also unveiled discrepancies in many datasets. Additionally, it enabled the augmentation of relevant information, such as the sex and life stage of the biological samples, which had been originally omitted in over 90% of cases. Our work not only enhances the quality and reliability of epigenomic data, but also underscores the value of harmonized metadata in ensuring the integrity of large-scale public datasets.

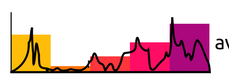
EpiATLAS



EpiATLAS: Harmonized data and metadata of 7.4k experiments conducted in 2216 human samples from IHEC

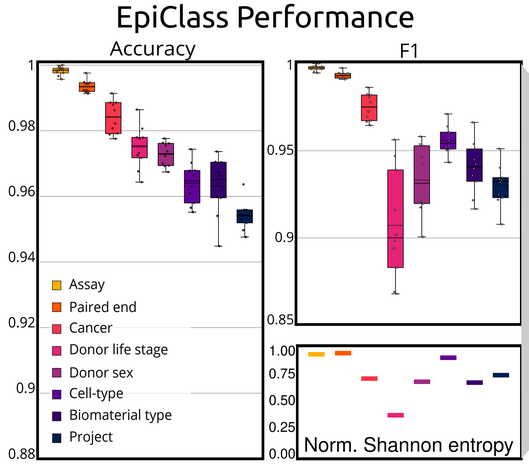
EpiClass: Set of neural network classifiers using EpiATLAS data and metadata for training



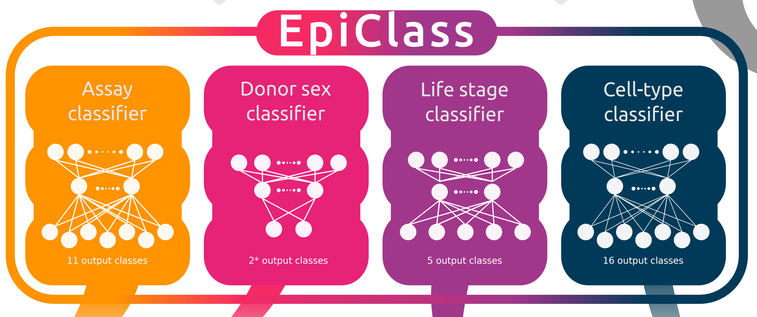


The signal is averaged in 100kb bins *(without chrY)*

EpiClass Performance



Stratified 10-fold Cross-Validation

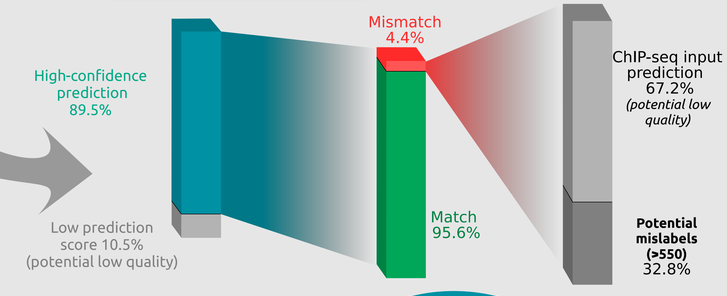


Public resources

ChIP-Atlas	40.2k datasets
ENCODE	4.3k datasets (not in EpiATLAS)

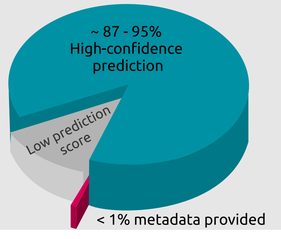
Uniformely reprocessed data with different pipelines

Prediction results



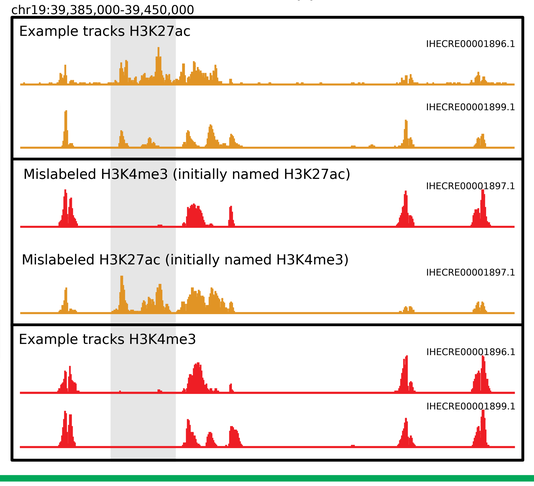
Metadata complementation

Sex
Life stage
Cancer

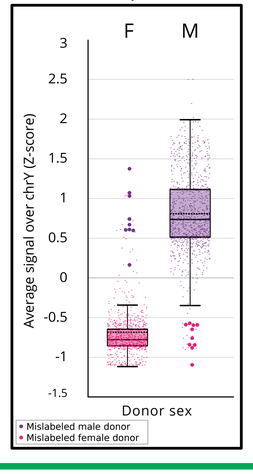


***Mislabels**: During the training of EpiClass, a few samples from EpiATLAS originally labeled incorrectly were discovered and corrected for Assays and Sex*

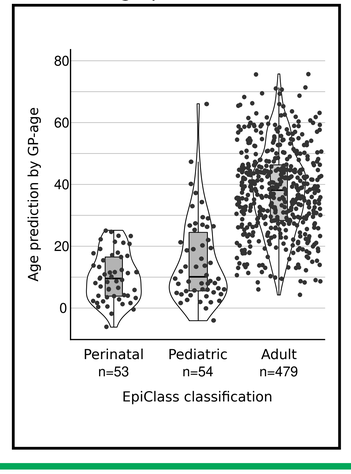
Identification of swapped datasets



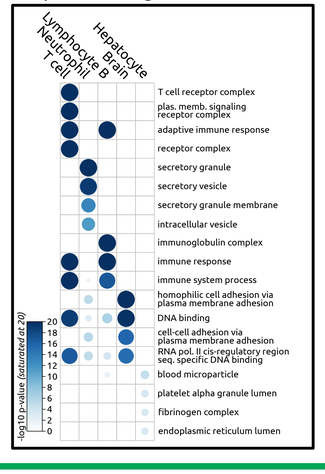
Support of Donor sex mislabeled prediction



Orthogonal analysis for age prediction



Gene ontology of cell-type important regions from SHAP



Perspectives

Train new models using all available ChIP-Seq datasets and publish an efficient tool that can be incorporated into usual pipelines to help identify problematic datasets and complement metadata

References

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